Nguyen Khoi Nguyen (Alan)

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Machine Learning Engineer | Computer Vision | Biomedical Imaging | Production ML Google Scholar | alanknguyen.com

Experience

Research Engineer · ***Boston University*** · *DCL & H2X Labs* *Apr 2025 – Present · Boston, MA*

- Full-time research on robot learning and machine perception with Prof. Wenchao Li (Dependable Computing Lab) and Prof. Eshed Ohn-Bar (H2X Lab): end-to-end autonomous driving in CARLA and improving the generalization and reliability of vision-language models (VLMs) and vision-language-action models (VLAs) across simulated and real-world robotic platforms.
- Built the 3D perception stack for a 7-DoF manipulator (camera calibration, point clouds, segmentation, grasp pose estimation, ROS1/ROS2, Gazebo/Isaac Sim); ran a 400-trial benchmark of vision-language models as task planners and reinforcement-learning benchmarking in simulation (Stable-Baselines3).

Principal Software Engineer · ***Crown Barber Studio*** · *3D Hair Scanning* *Oct 2025 – Apr 2026 · Contract, Remote*
App

- Sole engineer on a startup 3D hair-scanning product: built the entire iOS app (ARKit capture, neural 3D reconstruction, on-device rendering) and its cloud API (FastAPI, MongoDB, Firebase); shipped to the App Store, credited as developer.

Machine Learning Intern · ***Scale AI*** *Jun 2025 – Sep 2025 · San Francisco, CA*

- Developed AI training and evaluation procedures for computational physics (finite element, FDTD, beam propagation); built benchmarking suites in Python, C++, and MATLAB.

Undergraduate Researcher · ***Computational Imaging Systems Lab***, *BU* *Feb 2025 – Sep 2025 · Boston, MA*

- Researched intensity diffraction tomography with Prof. Lei Tian (3D refractive-index reconstruction from microscopy): transfer-function derivations, Python forward models, and Fourier-domain fidelity analysis.

Data Science Intern · ***Leuko*** · *Biomedical Optics & Machine Vision* *May 2023 – Aug 2023 · Boston, MA*

- Analyzed blood samples and developed OpenCV-based computer vision pipelines to detect and characterize blood cells from optical microscopy; built dataset versioning tooling for labeled biomedical image sets.

Publications

- J. Li, **N. Nguyen**, M. Lai, I. Ch. Paschalidis, J. H. Huggins, "Robust Model Selection for Discovery of Latent Mechanistic Processes," arXiv:2602.22062, 2026.
- **N. K. Nguyen**, "Quantum States of Light in Cavity QED: Wigner Function Dynamics and Atom–Field Entanglement in the Jaynes–Cummings Model," under review, *European Journal of Physics*, 2026.

Technical Skills

AI/ML & Gen AI	PyTorch, scikit-learn, Hugging Face Transformers, LLM/VLM-based pipelines and agents, reinforcement learning (Stable-Baselines3), evaluation design, statistical model selection
Computer Vision	OpenCV, Open3D, SAM, GroundingDINO, camera calibration, 3D reconstruction, 6-DoF pose estimation, segmentation, microscopy imaging
Cloud & MLOps	AWS, Docker, Linux, Git, CI/CD, FastAPI, React, PostgreSQL, MongoDB, Firebase; model validation, production monitoring
Languages & Robotics	Python, C++14/17, SQL, JavaScript, MATLAB, YAML; ROS1/ROS2, Gazebo, Isaac Sim, CARLA

Education

M.S. Electrical & Computer Engineering, Boston University — withdrew for a full-time research role *2025 – 2026*
B.A. Physics & Computer Science (Dual Major), Boston University *2020 – 2025*

Leadership & Distinctions

Co-Founder & President, **Boston University AI Society** (Jan 2023 – Sep 2025): led BU's largest student-run AI community; raised $10,000+ for AI and robotics hardware; advised BU Student Government on AI policy (TerrierGPT); taught BUAIS 185: Machine Vision; lead TA, EC 545 Cyber-Physical Systems.

Vietnamese Reserve National Team, 2018 International Mathematical Olympiad (IMO)